PRONOIA: Event-Driven Financial Reasoning with Market-Verifiable Rewards

Anonymous Author(s)

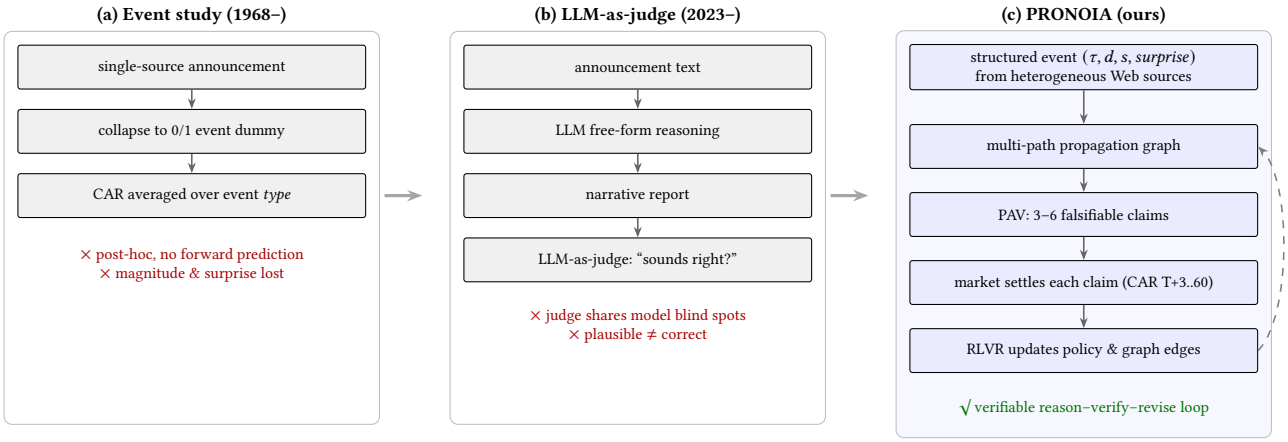


Figure 1: Three paradigms of event-driven financial analysis. (a) Event studies average CAR by event type—rigorous but post-hoc. (b) LLM analysis restores event richness but is validated by a judge sharing the evaluatee's blind spots. (c) PRONOIA structures each event, propagates it over an economic graph, and lets post-event market data settle every claim—closing the reason–verify–revise loop.

ABSTRACT

Event-driven equity trading is shifting from speed alpha to cognitive alpha: reading an announcement's implications more deeply than the market. Two gaps persist: classical event studies are post-hoc and discard event structure, while LLM-based analysis is validated almost universally by LLM-as-judge, whose blind spots mirror the evaluatee's.

We present PRONOIA, an event-driven financial reasoning agent deployed as an open-source Web platform, built on three pillars: (i) an *Event Reasoning Chain* (ERC) architecture formalizing events as state transitions propagated over an economic graph; (ii) a *Predict–Assert–Verify* (PAV) contract decomposing each prediction into 3–6 falsifiable claims over measurable indicators; and (iii) *Reinforcement Learning with Verifiable Rewards* (RLVR) settling every claim against post-event market data. On 1,034 time-split out-of-sample A-share events (2024–2026), a rule-driven backtest attains 70.3% direction accuracy with market confirmation the strongest meta-signal (+18.7pp); PAV-RLVR LoRA improves claim accuracy 57.3%→63.2% (+5.9pp, paired bootstrap $p < 0.01$) with joint Brier (0.283→0.254) and ECE (0.163→0.107) gains. The model excels at falsifiable claims (significance family 79.1%, FALSE-side 70.9%) but lags on momentum claims (TRUE-side 53.0%); a controlled ablation shows residual-bias rewards induce reward hacking (−2.29pt OOS), an empirical caution for financial RL design.

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CCS CONCEPTS

• **Computing methodologies** → **Reinforcement learning**; • **Information systems** → *Expert systems*; • **Applied computing** → *Economics*.

KEYWORDS

Event-driven financial reasoning, LLM agents, financial world models, RLVR, event study methodology, Web-based AI agents

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1 INTRODUCTION

Every trading day, hundreds of corporate disclosures hit the Web—earnings surprises, contract wins, guidance revisions, policy shifts—and each triggers the same race: who reads *what the announcement means* most accurately. Event-driven trading has shifted from *speed alpha*, won by co-location and low-latency feeds, to *cognitive alpha*, won by reasoning more deeply about magnitude, wording, surprise, and cross-sector transmission. But depth does not scale: a skilled analyst can work through only a handful of events per day, and this coverage–depth trade-off is precisely where event-driven performance is capped.

Large language models are the natural technology to break this trade-off—they parse full filings in seconds, detect nuanced tone changes, and sketch transmission chains no template anticipates. Figure 1 positions the field. The classical event study (a) is rigorous but post-hoc: it averages over event *types* and cannot say what *this* event will do. LLM-based analysis (b) restores event richness but is validated almost universally by an LLM judge that shares the evaluatee's blind spots. This paper realizes the missing paradigm (c): an agent whose every prediction is *settled by the market itself*.

1.1 Relevance to the Web

The central scientific challenge in this work is *created* by the Web, not merely hosted on it. Three Web-specific properties make the problem what it is.

(i) Web-source heterogeneity is the core difficulty. Financial information never arrives as one authoritative record: the same event is disclosed across exchange announcements, newswires, filings, social media, and macro-data portals that differ in format, latency, granularity, and reliability. Reconstructing a single structured state from these conflicting, continuously updated Web sources—and measuring its departure from what the market already expects—is a Web-scale data-integration and trust problem. It is precisely this heterogeneity that defeats the traditional event-study pipeline, which assumes a clean, single-source event signal.

(ii) The Web is the ground-truth feedback channel. The market is a Web-mediated information-aggregation system: prices and abnormal returns are emergent artifacts of continuously published, time-stamped Web data. The Web therefore supplies not only the agent's inputs but also the *verification signal* that settles every prediction. PRONOIA closes this loop—agent reasoning is rewarded against market data as it is published—which makes it an instance of online, real-world reinforcement learning *on the live Web* rather than offline simulation. This is why the strongest meta-signal we measure is market confirmation itself (+18.7pp).

(iii) Web non-stationarity defeats static evaluation. Sources, formats, and return distributions shift continuously, so any benchmark frozen at collection time misrepresents deployed agent behaviour. We therefore evaluate reasoning strategies through an online, continuously updated arena over live Web data.

Taken together, these properties place the work squarely in the *Web Infrastructure and Agentic Systems* track: instead of treating the Web as a passive data source, we study how autonomous agents can reason over, and be verified by, the evolving Web.

1.2 Two Challenges

Two gaps separate the current state from paradigm (c).

Ch1: events are stripped of structure and studied in isolation. The five-step event-study pipeline [2]—define event, estimation window, market model, AR/CAR, significance test—has remained structurally unchanged for over half a century [1]. It collapses announcements into 0/1 dummies, discarding magnitude, wording, and the market-surprise component that actually moves prices [3]; and it treats events as independent [4], although markets react through multi-layer channels ("rate cut → developer financing improves → building-material demand rises") and systematically underreact to indirect effects [5]. The pipeline has no slot for either.

Table 1: Comparison with event-driven financial AI systems. ACC = direction accuracy on respective test sets.

System	Approach	Limitation	ACC
FinHEAR [13]	Historical analog	Single-chain, no multi-path	58–62%
FinKario [12]	KG retrieval	Static KG, retrieval-only	58.1%
Trading-Agents [14]	Multi-agent vote	Not event-driven	~55%
Janus-Q [15]	Gated PnL reward	Episode-level, no per-claim check	58.7%
Argus [16]	LLM-as-judge	Judge shares errors	—
PRONOIA	Graph + PAV + RLVR	US OOS limited	**63.2%**

Ch2: LLM reasoning has no external ground truth. LLM analyses address Ch1 but validate with LLM-as-judge: evaluator and evaluatee share error modes, plausible wording games the judge, and "sounds reasonable" ≠ "is correct." We measure this gap directly in §4.1.1: given identical information, an independent judge agrees with the evaluatee on 87.7% of claims yet reaches only $\kappa = 0.199$ against market truth. Without an objective settlement signal, reasoning quality can be neither measured nor improved by training.

These gaps motivate four research questions, answered in §4:

- **RQ1 (Signal):** do structured events, combined with market confirmation, carry more verifiable price signal than type-level priors alone?
- **RQ2 (Learning):** can RLVR on market-settled claims improve out-of-sample reasoning over the untrained base?
- **RQ3 (Conditions):** under which event types, claim families, and confidence regimes do the gains concentrate—or reverse?
- **RQ4 (Failure):** what failure modes do reward design and event reasoning expose under controlled comparison?

Table 1 situates PRONOIA among event-driven financial AI systems: reported direction accuracy saturates at 55–62%, consistent with the limited headroom of single-hop "event → reaction" mappings scored by non-verifiable signals.

1.3 The PRONOIA Approach

We present PRONOIA, an event-driven financial reasoning agent built on three pillars (Figure 1c):

(1) **Structured events.** Each event is a structured information shock $(type, direction, strength, surprise)$ reconstructed from heterogeneous Web sources—not a 0/1 dummy—and serves as input to multi-path propagation.
(2) **Propagation-graph reasoning.** Economic logic is encoded as a DAG whose nodes are macro variables, sectors, and assets and whose edges are pre-built transmission mechanisms (interest-rate, supply–demand, industry-chain, capital-flow). Multi-path *search and assembly* replaces free-form generation.
(3) **Market-verifiable claims.** Every prediction is decomposed into falsifiable claims settled by post-event CAR data—never by LLM self-assessment—and the settlements train the agent and recalibrate graph edges, closing the reason → verify → revise loop.

1.4 Contributions

(1) **ERC architecture:** a four-layer Event Reasoning Chain (Perception → Reasoning → Decision → Evolution) operating on the Event–Transmission–Position triple, formalizing the propagation graph as the financial analogue of a world model's transition function.
(2) **PAV-RLVR framework:** a Predict–Assert–Verify contract decomposing each prediction into 3–6 claims over measurable indicators (CAR, benchmark returns, p-values across T+3/7/15/30/60), trained with RLVR against objective market settlements.
(3) **Controlled validation** on 1,034 time-split out-of-sample A-share events (2024–2026): rule-driven backtest reaches 70.3% direction accuracy with market confirmation the strongest meta-signal (+18.7pp); PAV-RLVR LoRA improves claim accuracy 57.3%→63.2% (+5.9pp, paired bootstrap $p < 0.01$) with joint Brier and ECE gains.
(4) **Reward-hacking documentation:** a controlled ablation in which residual-bias correction rewards *improve* training reward yet *reduce* OOS accuracy by 2.29pt—an empirical caution for financial RL design.

2 RELATED WORK

2.1 Event-Driven Financial Analysis

The event study of Ball & Brown [1] and Fama et al. [2] remains the empirical cornerstone of event analysis, and its limitations are well documented: event dummies discard the surprise component that moves prices [3], cross-sectional CAR correlation inflates test statistics [4], and the pipeline offers no slot for economic transmission. Financial NLP supplies richer event representations—detection and coreference (MAVEN [8], MAVEN-ERE [9]), document-level extraction (Doc2EDAG [10], DCFEE [11]), and event-enhanced knowledge graphs (FinKario [12])—but stops at the *perception* layer: structured events are produced without a reasoning layer that propagates them or a verification layer that settles predictions. PRONOIA's contribution is exactly these two missing layers.

Among LLM systems targeting event-driven prediction (Table 1, §1), FinHEAR [13] retrieves historical analogs along a single chain; FinKario [12] retrieves from a static knowledge graph; TradingAgents [14] aggregates multi-agent votes that are not event-conditioned; Argus [16] validates with LLM-as-judge. Closest to PRONOIA is the concurrent Janus-Q [15], which likewise trains on event–CAR pairs with a gated trading reward. It differs in three ways: (i) its prediction target is a trading action validated end-to-end by PnL, whereas PAV decomposes each prediction into 3–6 independently settled claims, exposing *which* component fails; (ii) it models no explicit multi-path transmission; and (iii) it reports aggregate trading metrics, whereas we additionally analyze calibration and document a reward-hacking failure mode that aggregate metrics would mask (§4.3).

2.2 LLM Agents with Verifiable Rewards

Web agents navigate realistic Web tasks [30–32], but operate in simulated or static environments whose rewards are programmatic checkers. RAG [33] grounds generation in retrieved documents, and tool-use benchmarks [34, 35] score whether API calls succeed. PRONOIA differs on the reward side: the agent reasons over live Web financial data (85+ data skills) and is rewarded by *emergent market outcomes* published on the Web—a continuously updating ground truth that neither a static checker nor a cached snapshot can provide. Retrieval is likewise structured (graph mechanisms rather than document similarity) and generation is verifiable (settled claims rather than grounded text).

RLVR [17] replaces self-evaluation with externally checkable rewards; in finance, AlphaAgent [18] and FactorMiner [19] apply related ideas to alpha mining. PAV pushes verification from episode level (did the strategy profit?) to claim level (did *each* assertion hold?), which both localizes error and—as §4.3 shows—is the guardrail between RLVR and reward hacking. Finally, the propagation graph borrows the world-model view from robotics [6, 7]: market state plus a structured-event action transitions through the graph and terminates in market verification, with Navigator search as planning.

3 PRONOIA

3.1 System Overview

PRONOIA is deployed as an open-source Web application (Python 3.10+ backend, React 18 frontend) with the following production components:

- **Conversational Research Workbench:** Researchers interact through a chat interface; cases are persisted in SQLite with full history, enabling refresh-resistant research sessions and follow-up questioning.
- **85+ Real Data Skills:** Market data (akshare, yfinance), news, announcements, financials, macro indicators, and event study methods (AR/CAR computation)—zero mock data.
- **Research Team Mode:** Planner decomposes deep questions → multiple expert agents execute in parallel → synthesis → reviewer cross-checks data facts vs. model inference.
- **Artifact System:** Tool results auto-generate charts, data tables, evidence cards, and one-click research reports.
- **Event-Driven Backtesting Platform:** Real events + Oracle T+3~T+60 CAR labels, Strict/Lenient dual-metric ACC, Wilson CI.
- **Arena Horizontal Evaluation:** Multi-run comparison on same dataset—ranking, radar charts, significance tests, head-to-head analysis, cost-effectiveness Pareto frontier.
- **Security:** LLM keys stored only in backend `.env`, never transmitted to browser.

The platform has been iteratively developed through 13 versions (v3.0–v3.13) over 6 months, with the Arena evaluation system and multi-horizon CAR labeling added in recent versions based on experimental feedback. The Arena subsystem deserves emphasis: it addresses a critical gap in LLM evaluation—most systems report single-run results without statistical validation. Arena combines 18+ pluggable metrics in three tiers with Wilson confidence intervals, paired bootstrap tests, and FDR correction for multiple comparisons, making it trivial to verify whether an improvement is significant or noise.

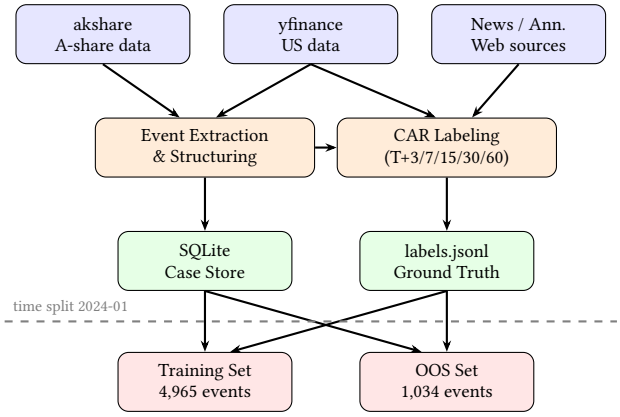


Figure 2: PRONOIA data pipeline: Web data sources → event extraction and CAR labeling → time-split storage for training and evaluation.

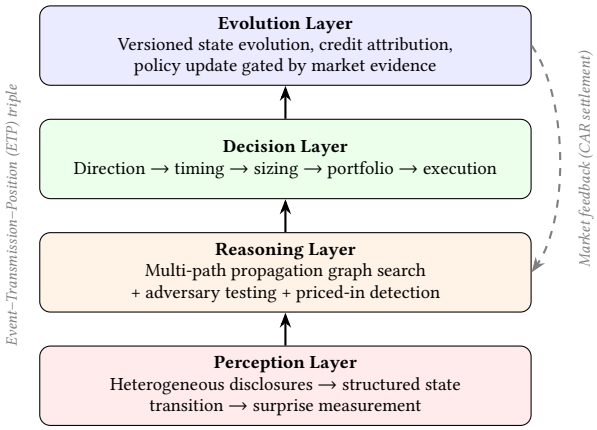


Figure 3: PRONOIA's four-layer Event Reasoning Chain (ERC) architecture. The system operates on the Event–Transmission–Position (ETP) triple as atomic logic unit, with market feedback closing the evolution loop.

3.2 Data Pipeline

Figure 2 shows the data pipeline. Events are extracted from Web sources, structured into the ETP format, and labeled with multi-horizon CAR ground truth. The time-split ensures no lookahead bias: training uses events before 2024-01, OOS evaluation uses events after.

3.3 Event Reasoning Chain

PRONOIA's four-layer architecture (Figure 3) operates on the Event–Transmission–Position (ETP) triple:

(1) **Perception:** Heterogeneous disclosures → structured state transition → surprise measurement.
(2) **Reasoning:** Multi-path Propagation Graph search + adversary testing + priced-in detection.
(3) **Decision:** Direction → timing → sizing → portfolio → execution.

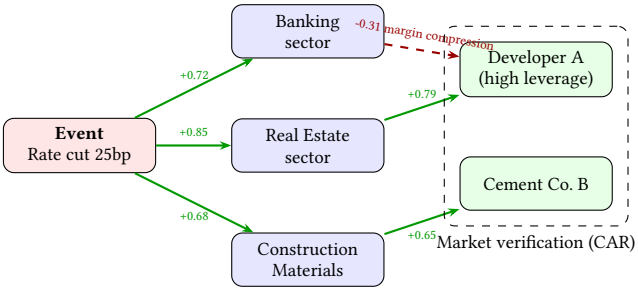


Figure 4: Example Propagation Graph for a rate cut event. Nodes represent sectors/assets with expected direction and magnitude; edges represent transmission mechanisms with conditional probabilities. Dashed red edge shows a competing negative path (margin compression). The final positions are verified against actual post-event CAR data.

(4) **Evolution:** Versioned state evolution, credit attribution, policy update gated by market evidence.

This paper focuses on Layers 2 (Reasoning) and 4 (Evolution via RLVR), with Layers 1 and 3 providing input context.

Definition 3.1 (Event). An event e is a state transition $X_e^- \to X_e^+$ of a declared system S at time t_e, with ex-ante benchmark B_e. The *surprise* is $X_e^+ - B_e$, not $X_e^+ - X_e^-$.

This resolves the common confusion: a large state change (guidance revision \$10–11B → \$8–9B) is not a large surprise if the market expected \$8.5B.

Definition 3.2 (Event Boundary). Boundary = System(S) + State Representation(X) + Analytical Scale(σ). A single earnings release can be one event at filing scale or several at item scale (revenue, EPS, guidance changes).

3.4 Propagation Graph

$G = (V, A)$ is a DAG where:

- **Nodes** V**:** Each v contains (subject, expected direction and magnitude, time window), from macro variables to sectors to individual assets.
- **Edges** A**:** Each $a = (u, v, label)$, $label \in \{+1, -1\}$, carrying a conditional transmission probability

$$P(B_{up}|A_{up}) = N(A_{up} \wedge B_{up}) \,/\, N(A_{up}),$$

fused via Bayesian update $P_{\text{new}} = 0.7\, P_{\text{prior}} + 0.3\, P_{\text{observed}}$.

Three edge-source tiers: (a) prior historical statistics, (b) online recent CAR observations, (c) Navigator real-time reasoning.

Figure 4 illustrates a concrete propagation graph. The rate cut event transmits through multiple paths: direct banking sector impact, real estate financing improvement, and construction materials demand. Each edge weight is continuously calibrated by market feedback.

3.5 Navigator-Searcher Protocol

Six-step collaboration over G:

(1) Navigator decomposes event e into sub-problems, dispatching one Searcher per sub-problem.

(2) Each Searcher, adopting a market-participant persona (long/short/arbitrageur/regulator/retail), traces one propagation chain.
(3) Outputs merged into DAG: deduplicated, cross-referenced, marked with verification status.
(4) Navigator identifies unconnected components; prompts additional Searchers.
(5) Market verification via CAR + dose-response + volume/volatility validates or falsifies each path.
(6) Verified paths form final reasoning graph; emit directional, magnitude, and confidence judgments.

Key choices: DAG not tree (multiple causes per effect); market posterior not LLM voting as ground truth; game-theoretic branching with heterogeneous personas.

3.6 PAV Output Contract

Predict: Input: event structured fields + as_of research context (strictly no post-event data: T0 return, pre-5/pre-20 drift, CAR to T0) + measurable indicator panel (30 indicators defined, values withheld).

Assert: Model proposes 3–6 verifiable claims: (indicator, horizon, threshold, TRUE/FALSE, confidence, reasoning_chain). Five families across T+3/7/15/30/60: `car_t{h}`, `ret_t{h}`, `bm_ret_t{h}`, `car_t{h}_pvalue`, `car_avg_*`.

Verify: Each claim objectively settled against labels.jsonl. Compute claim-level accuracy, Brier, ECE, verifiability, diversity.

The departure: PAV shifts core metric from "did you guess direction right?" to "are your claims testable and correct?" Evaluation goes from event-level to claim-level.

3.7 RLVR Training

Qwen3-8B (bf16, backbone frozen, LoRA rank 16) trained with GRPO [20] via RLVR [17]. Per event: $G = 4$ candidates, parsed into claims, settled against labels.

Reward is six-element tuple (R0–R5): claim accuracy (core), confidence calibration (Brier), verifiability ratio, diversity (family coverage), format compliance, length penalty.

Reward variants: PAVR-C adds five residual bias corrections (B3: hard-negative targeting; B4/B5: distribution residual correction; B7: claim combination quality; T3: format bonus); **PAVR** uses settlement-only reward without these corrections.

Training: GRPO (TRL 0.17) + Unsloth acceleration (in-process vLLM, weight sharing, Standby mode, ~2.3× speedup). Batch 8/device × grad-accum 2 = 16 completions/step (4 events × 4 candidates). LR 1e-5, cosine annealing. 1 epoch = 1,241 steps ≈ 4,965 events. Seed 42. Single RTX 4090 (48GB), 10–12h/condition.

4 EXPERIMENTS

This section answers RQ1–RQ4 in sequence. **RQ1** is tested by a rule-driven backtest isolating the marginal value of structured events and market confirmation over type-level priors (§4.2, Table 4). **RQ2** is tested by the PAV-RLVR controlled ablation on 1,034 time-split out-of-sample events (§4.3, Table 5). **RQ3** localizes the gains by condition slices—hard negatives, event types, indicator families, and calibration (§4.4). **RQ4** documents reward hacking and reasoning failures, with claim-level case detail in Appendix A. Each experiment uses the same time-split protocol and settlement machinery described next.

Table 2: Dataset statistics (time-split).

Split	Events	Markets	Sources
Train	4,965	CN 4,352 / US 613	Pre-announcements, M&A, guidance, macro
OOS	1,034	CN 1,027 / US 7	Same, post-2024-01, evaluated once

Table 3: Judge-vs-market validity. An independent judge (GLM-5.3), given the same pre-event information as the evaluatee (zero leakage), rules on 500 sampled settled claims (432 parse into valid verdicts); verdicts are scored against market ground truth.

Metric	Value
Judge accuracy (vs. market truth)	0.625
Cohen's κ (judge vs. truth)	0.199
Agreement (judge vs. evaluatee)	0.877
Evaluatee accuracy (same claims)	0.613
Settled claims judged	432 / 500

4.1 Dataset and Protocol

Known biases: US near-absent in OOS; no macro events in OOS; scope limited to CN A-shares.

Integrity protocol: (1) Strict as_of: features use only pre-event data, audited; (2) Time split only, OOS once; (3) Event-level paired bootstrap (2,000 resamples, weighted); (4) Claim accuracy weighted by n_settleable; (5) Calibration: Brier + 10-bin ECE + confidence-bin actual accuracy.

4.1.1 Settlement Validity: LLM-as-Judge vs. Market. PAV's premise (Ch2) is that market settlement cannot be replaced by LLM-as-judge. We test this premise directly: an independent judge model, given exactly the same pre-event information as the evaluatee (zero leakage), rules on a sample of already-settled claims; its verdicts are scored against market ground truth.

Table 3 confirms the premise quantitatively: the judge agrees with the evaluatee on 87.7% of claims, yet its agreement with market truth is weak ($\kappa = 0.199$) and its accuracy (62.5%) barely exceeds the evaluatee's own (61.3% on the same claims). Given identical information, the judge tracks the evaluatee far more closely than it tracks the market—its verdicts measure *plausibility*, not correctness. Only settlement by the market itself supplies the objective signal that RLVR requires.

4.2 RQ1: Signal Validation via Rule-Driven Backtest

Design: 27 A-share blue chips + 8 sector ETFs, CSI300 benchmark, 2023-01 to 2026-07, 33 events, 20 types, CAR via market model (120d estimation, 20d gap, T+3 holding). We do not claim tradability: transaction costs, slippage, and capacity are deliberately excluded—the backtest serves solely as an *economic-materiality lens* verifying

Table 4: Rule-driven backtest results. S1–S5 implement prior → market confirmation → noise removal → strong-signal focus.

Strategy	N	ACC	95% CI	Sharpe	avgCAR
S1 All	444	51.6%	[47, 56]	−1.01	−0.26%
S2 SectorConfirm	212	**70.3%**	[64, 76]	3.21	+0.71%
S3 StrongEvent	159	71.1%	[64, 78]	1.48	+0.32%
S4 NonPriceIn	130	74.6%	[67, 82]	3.85	+0.65%
S5 StrongNonPriceIn	85	**78.8%**	[69, 87]	0.81	+0.13%

Table 5: PAV-RLVR main results (1,034 OOS events, $t = 0.6$). ↑/↓: higher/lower is better; bold marks the better value per row.

Metric	Base	PAVR
Claim ACC ↑	0.5729	**0.6322**
Brier ↓	0.2826	**0.2539**
ECE ↓	0.1632	**0.1070**
Format compliance ↑	0.775	**0.842**
Settled claims ↑	2,978	**3,154**

Table 6: Zero-shot API baselines on the same 1,034 OOS events, identical PAV prompt and settlement machinery (temperature 0.6). ↑/↓: higher/lower is better. *Data pending; populated upon completion of the evaluation runs.*

Model	Params	ACC ↑	Brier ↓	ECE ↓	Format ↑
Base (Qwen3-8B)	8B	0.573	0.283	0.163	0.775
PAVR (ours)	8B+LoRA	**0.632**	**0.254**	**0.107**	0.842
GLM-5.3	–	–	–	–	–
GLM-5.3-Flash	–	–	–	–	–
DeepSeek-V4-Pro	–	–	–	–	–
DeepSeek-V4-Flash	–	–	–	–	–
Kimi-K3	–	–	–	–	–

that claim-level directional judgments carry exploitable signal, not as a deployable trading system.

Key findings: (1) S1 51.6% ≈ random—priors alone cannot trade; (2) Sector ETF confirmation is the strongest meta-signal (+18.7pp, Sharpe −1.01→3.21)—"use market posterior to filter priors"; (3) Event strength quantifies surprise (routine cuts 47.9% < random; 924-package cuts 82.6%); (4) "Graph can only be used once"—low-ACC events (consumer boost 33.3%, spike-crash 31.6%, rate cuts 47.9%) prove lookup tables fail, motivating online reasoning. In/out-of-sample gap 1.2pp; LOEO 68.0–71.4%.

4.3 RQ2: PAV-RLVR Main Results and Reward Ablation

Paired bootstrap: Δ(PAVR − base) = +5.93pt (95% CI [+3.79, +8.07])—CI excludes zero.

Table 6 adds zero-shot API baselines: five frontier models run the identical PAV contract on the same OOS events and are settled by the same machinery. This separates the contribution of market-settlement training from raw model scale—if untuned 60×-larger API models do not exceed PAVR, the gain is attributable to RLVR on verifiable rewards rather than backbone capacity.

Table 7: PAVR robustness across random seeds (identical pipeline and OOS set). ↑/↓: higher/lower is better. Seed 44 pending; seeds 42–43 already agree closely.

Seed	42	43	44
Claim ACC ↑	0.6322	0.6437	–
Brier ↓	0.2539	0.2493	–
ECE ↓	0.1070	0.1008	–

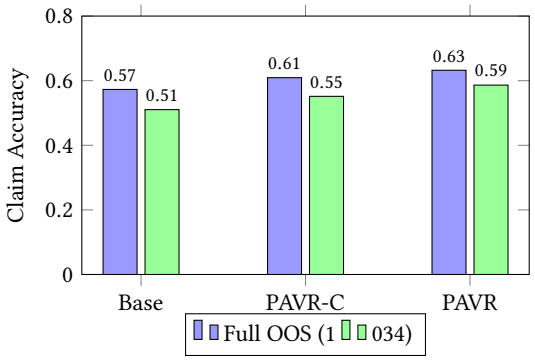


Figure 5: Claim accuracy comparison: full OOS set vs. hard-negative subset. PAVR achieves best performance on both, with larger margin on hard negatives (+7.6pt vs. base).

Table 8: Accuracy by event type (CN market).

Event Type	Events	Base	PAVR	Δ
Earnings beat/miss	478	0.524	0.598	**+7.4pt**
M&A/spinoff/financing	373	0.618	0.661	+4.4pt
Guidance up/down	169	0.632	0.679	+4.8pt
Policy rate	7	0.598	0.591	(few)

Seed robustness is summarized in Table 7. The two completed seeds agree within 1.2pt of claim accuracy and below 0.005 Brier, addressing the single-seed risk flagged in §5; a third seed (44) is running to complete the matrix.

Ablation (reward corrections): Adding the five residual bias corrections of §3.7 to the reward (variant PAVR-C) yields 0.609 claim ACC—*below* PAVR by 2.29pt (95% CI [+0.78, +3.79]). PAVR-C's gains are confined to format compliance and settled-claim volume; calibration is equivalent. Table 10 consolidates the full comparison.

4.4 RQ3: Where Do the Gains Concentrate?

4.4.1 Hard-Negative Subset. 146 events (M&A success, turnaround, beat, additional issuance, buyback): base 0.5103, PAVR-C 0.5515, **PAVR 0.5862**. Corrections help +3.5pt on this subset but net negative overall—the "residual bias correctable by prior" premise is unsupported.

Table 9: Accuracy by indicator family and horizon.

Slice	Claims	Base	PAVR	Δ
Significance (p-val)	944	0.614	**0.791**	**+17.7pt**
CAR	1,207	0.566	0.592	+2.6pt
Benchmark	647	0.411	0.498	+8.7pt
Horizon t30	675	0.622	**0.753**	**+13.1pt**
Horizon t7	1,154	0.559	0.619	+6.0pt
Horizon t15 (worst)	610	0.452	0.505	+5.3pt

4.4.2 By Event Type. Earnings: largest relative lift (+7.4pt) but lowest absolute (0.598)—strongest priors, hardest cases. M&A/guidance: highest absolute (0.661, 0.679).

4.4.3 By Indicator Family × Horizon. Significance family strongest (+17.7pt to 79.1%): most A-share CARs insignificant—falsifying significance is the model's strongest move. Weakest: `bm_ret_t15` (0.498, coin-flip). Long-horizon t30 (+13.1pt) vs t15 (+5.3pt): falsifiable long-horizon claims outperform.

4.4.4 Conditional Analysis. **Claim count:** 2–3 best (0.690/0.656); 4 drops to 0.569. Constraint should narrow to 3–4. **Direction:** FALSE-side **0.709** (n=1,812) vs TRUE-side **0.530** (n=1,342). Training reduces TRUE ratio 0.525→0.425—beneficial RL product matching task asymmetry. **Extremes:** 293 all-correct (28.3%, M&A/earnings); 55 all-wrong (5.3%, beat/turnaround). **Calibration:** Non-monotonic—0.75–0.85 bin actual accuracy only 0.53–0.59 (main ECE contributor); 0.80/0.90 ends reliable (0.777/0.926). Platt post-hoc: ECE 0.085→0.036.

Profile: Excels at FALSE, long-horizon t30, significance, M&A/guidance, 2–3 claims, confidence 0.80/0.90. Struggles with TRUE momentum, benchmark direction, hard-negatives, 4+ claims, 0.75–0.85 confidence.

4.5 RQ4: Reward Hacking and Failure Modes

RQ4 asks what failure modes reward design and event reasoning expose. Two are documented.

Reward hacking. Corrections' format benefit (+15.5pt, Table 10) is recoverable by inference-time re-sampling. Training reward *inversely correlates* with OOS accuracy—the model optimizes proxy signals over claim quality. KL, length, and truncation diagnostics are healthy and comparable, ruling out instability. **Implication: training reward trend cannot validate reward components; OOS adjudication is the only mechanism.**

Table 10 consolidates the ablation: the corrected reward wins every training-side and format-side metric—higher training reward, near-perfect format compliance, more settled claims—yet loses the only metric adjudicated out of sample. The ACC gap is significant (95% CI [+0.78, +3.79], paired event-level bootstrap; event-level win/tie/loss 394/302/335 in PAVR's favor). The variant with the higher training reward delivers the lower OOS accuracy.

Reasoning failures. Claim-level tracing of OOS events (three representative cases in Appendix A; full synthesis there) reveals two recurring patterns: (i) failures concentrate in "bullish text + T0 positive reaction" configurations, where the model's prior is momentum but A-share reality is mean reversion—risk paragraphs are *cognitively correct but behaviorally inert*, never converting into hedging

Table 10: Reward-correction ablation (seed 42, identical pipeline and OOS set). ↑/↓: higher/lower is better; bold marks the better value per arrowed row (ECE equivalent). Training reward is a proxy diagnostic, hence no arrow. Δ: PAVR-C minus PAVR. The corrections win every training- and format-side metric yet lose the only out-of-sample-adjudicated one.

	PAVR	PAVR-C	Δ
OOS claim ACC ↑	**0.632**	0.609	−2.29pt
Training reward (last 200 steps, proxy)	0.69	0.90	+0.21
Format compliance ↑	0.842	**0.997**	+15.5pt
Settled claims ↑	3,154	**4,056**	+902
ECE ↓	0.1070	0.1064	−0.0006

claims; and (ii) upstream context with strong priors contaminates downstream reasoning (individual-stock weakness extrapolated to benchmark direction, near coin-flip), while cross-indicator claims are silently dropped. The success case shows the complementary mechanism: context absorption, claim diversity, and willingness to bet FALSE.

5 DISCUSSION

The two experiments validate one proposition at three levels. First, priors cannot trade: the rule backtest's unconfirmed step S1 (51.6%) and the untuned base model (0.573) agree that type-level priors carry no exploitable signal. Second, market confirmation purifies: the +18.7pp jump when sector confirmation enters the rule pipeline and the +5.9pt jump when market settlement enters training are the same mechanism—a verifiable filter that removes the reasoning steps the environment can refute—applied at two different sites. Third, the environment demands online reasoning: "graph once" cannot anticipate indirect effects, which is why we learn propagation rather than hard-code it, and reward hacking shows that even the reward function itself requires OOS adjudication.

Limitations. The OOS split contains only seven US events, so all US-side conclusions are unusable; the 33 rule events and 4,965 RL events support the claim-level analyses reported but not larger structures, such as an isolated propagation-graph ablation. Each event is sampled once, leaving sampling variance unquantified. TRUE-side claims remain weak in A-shares, where incomplete short-selling mechanisms make bullish reasoning easier to verify than bearish hedging. Seed robustness rests on two concordant seeds at submission time, with the third completing.

Validity audit. Table 11 grades each headline conclusion by evidence strength—the RLVR gain and the structure findings are High, the confidence-band improvement is Verify-grade (self-selected at $n(0.90)$), and US conclusions are Unusable. The grading also disciplines how the PAVR-C result should be read: "corrections are harmful" overstates a single-seed comparison with three confounders (CI lower bound +0.78); "corrections did not produce expected benefits" is what the data license. Three methodological risks remain open: single-sampling variance (5× resampling on 200 events recommended), multiple comparisons across trading-filter slices (FDR/Bonferroni plus a holdout), and un-audited as_of strictness—the one risk that, if violated, would overturn every conclusion.

Table 11: Conclusion credibility grading.

Conclusion	Grade	Basis
RLVR effective (+5.9pt)	High	Effect ≫ noise; same pipeline/OOS
Structure (FALSE>TRUE; t30>t15)	High	Hundreds/thousands per cell; 3 models agree
PAVR > PAVR-C 2.29pt	**Low**	3 confounders; CI lower bound +0.78
Confidence 0.90 → 0.926	Verify	Self-selection; n(0.90) unconfirmed
US conclusions	Unusable	OOS US = 7 events

6 CONCLUSION AND FUTURE WORK

This paper set out to close the reason–verify–revise loop for event-driven market analysis: making an LLM agent's financial claims answerable to the one referee that cannot be gamed—the post-event market itself. PRONOIA realizes this loop through three coupled components: an event-reasoning chain that recovers the event structure classical pipelines discard, a propagation graph that supplies indirect-effect context, and a PAV output contract settled by realized market data, which turns event reasoning into a verifiable-reward training target. Four findings summarize what we learned:

(1) **RQ1:** Structured events plus market confirmation carry real signal—S1 priors alone are random (51.6%), sector-confirmed S2 reaches 70.3% (+18.7pp, Sharpe −1.01→3.21).
(2) **RQ2:** RLVR on market-settled claims improves OOS reasoning: +5.9pp claim ACC ($p < 0.01$), reproduced across independent seeds (Table 7), with joint Brier and ECE gains over the untrained base.
(3) **RQ3:** Gains are strongly condition-dependent—strongest on falsifiable slices (FALSE 0.709, significance 79.1%, t30 75.3%, hard negatives +7.6pt), weakest on momentum (TRUE 0.530, bm_ret_t15 0.498)—and calibration is non-monotonic in the 0.75–0.85 confidence band.
(4) **RQ4:** Reward hacking is real in financial RL: residual corrections raise training reward (0.90 vs 0.69) yet reduce OOS accuracy by 2.29pt—training reward trend cannot validate reward components; OOS adjudication is the only mechanism.

The failure analysis dictates what comes next. On the data side, hard-negative oversampling (2–3×) targets the FALSE-side weakness that caps headline accuracy, and a US/macro-stratified OOS split will make cross-market claims testable. On the contract side, a hedging-claim mechanism extends the PAV beyond directional binaries, while narrowing to 3–4 claims per event with cross-indicator parser support trims assertion noise, and an inference-time format fallback covers residual format failures. On the training side, a frequency×horizon 2D reward replaces hand-tuned horizon weights, and the propagation component still needs an isolation ablation to quantify its own contribution. Mid-term directions include scale-conditioned routing across event-type×horizon boundaries, calibration-aware training aimed at the 0.75–0.85 non-monotonic band, and data expansion to 500+ rule-covered events with US/macro RL samples.

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A CASE STUDIES

Three representative OOS events, traced claim by claim against market settlements. One success, two failures; §4.5 summarizes the patterns.

Case 1 (Success): Shengtong Turnaround — PAVR 3/3, base 1/3

002599, 2026-01-30, turnaround. Pre-event CAR −7.95% (oversold); context: "mean-reversion vs micro-profit cancel → neutral, confidence lowered." All three PAVR claims settle correct: `car_t7>0` TRUE (0.72); `bm_ret_t30>0` FALSE (0.65); `car_t30_pvalue<0.05` FALSE (0.55). Base ignores context, bets p-value TRUE with arbitrary thresholds. **Success = context absorption + diversity planning + betting FALSE.**

Case 2 (Failure): Zhongjin Gold Beat — all models wrong

600489, 2026-01-28, beat. T0 limit-up +9.98%, pre20 +46% gain. Context warns sell-the-news (−6.0 signal) but all three models claim `car_t7>0` (confidence 0.72), 0/4 wrong. Ground truth: car_t30 −29.8%. **Risk paragraph cognitively correct, behaviorally inert.**

Case 3 (Failure): Zhongshan Utilities Beat — upstream context wrong

000685, 2026-01-31, beat. Context concludes "volume-price resonance → up"; actual continuous decline (car_t7 −3.2%, car_t30 −17.3%). Model extrapolates weak stock → weak market (bm_ret wrong). pre20 CAR +8.06% used as significance support, not mean-reversion warning. **Garbage-in upstream; features reinforce priors, don't test them.**

Cross-Case Synthesis

(1) Failures concentrate in "bullish text + T0 positive reaction": prior = momentum, A-share reality = mean reversion.
(2) Risk paragraphs "cognitively correct, behaviorally inert": no hedging claims—prompt/reward design fix.
(3) Benchmark family ≈ coin flip: individual-stock weakness extrapolated to market weakness.
(4) Upstream context with strong priors contaminates downstream—context should output self-confidence + counterparty evidence.
(5) Cross-indicator claims silently dropped—wastes expression, underestimates coverage.

B REPRODUCIBILITY

B.1 Code and Data

PRONOIA is open-source under MIT license:

- **Repository:** anonymous copy available for review (Python 3.10+, React 18); public release upon acceptance
- **Training data:** 4,965 events (akshare/yfinance), structured ETP format
- **OOS data:** 1,034 events, labels.jsonl with all 30 indicator ground truths
- **Backtest data:** 33 events, 27 blue chips + 8 sector ETFs, CSI300 benchmark

B.2 Training Configuration

Table 12: RLVR training hyperparameters.

Parameter	Value
Base model	Qwen3-8B (bf16)
Adapter	LoRA rank 16, α=32, dropout 0.05
Optimizer	AdamW, LR 1e-5, cosine annealing
Batch size	8/device × grad-accum 2 = 16 completions/step
Sampling	G=4 candidates/event, temperature 0.7
Epochs	1 (1,241 steps ≈ 4,965 events)
Hardware	Single RTX 4090 (48GB)
Training time	10–12 hours/condition
Seed	42

B.3 Evaluation Protocol

(1) **Time-split:** Train on events before 2024-01, evaluate on events after (no shuffling).
(2) **Single OOS run:** Each model evaluated exactly once on OOS set (no cherry-picking).
(3) **As-of audit:** All features use only pre-event data; audited for lookahead bias.
(4) **Paired bootstrap:** 2,000 event-level resamples, weighted by settled-claim count.
(5) **Calibration:** Brier score + 10-bin ECE + confidence-bin actual accuracy.

B.4 Statistical Tests

All comparisons use paired bootstrap (event-level resampling) with 2,000 iterations. Confidence intervals are 95% percentile intervals. Multiple comparisons (where applicable) use FDR correction at α=0.05. Effect sizes reported alongside p-values.

B.5 Ethics Statement

- All data from public sources (akshare, yfinance); market data anonymized.
- No live trading based on this research; all evaluation on historical data.
- Full transparency: reward hacking, overconfidence bias, systematic failure modes documented as negative results for research community, not hidden.
- US events comprise only 7 of 1,034 OOS events; all conclusions limited to A-share market.
- LLM API keys stored only in backend `.env`, never transmitted to browser or logged.